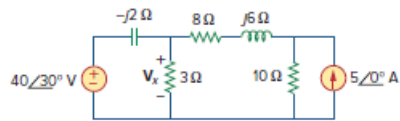


Problem

Determine V_x in the circuit of Fig. 10.62 using any method of your choice.

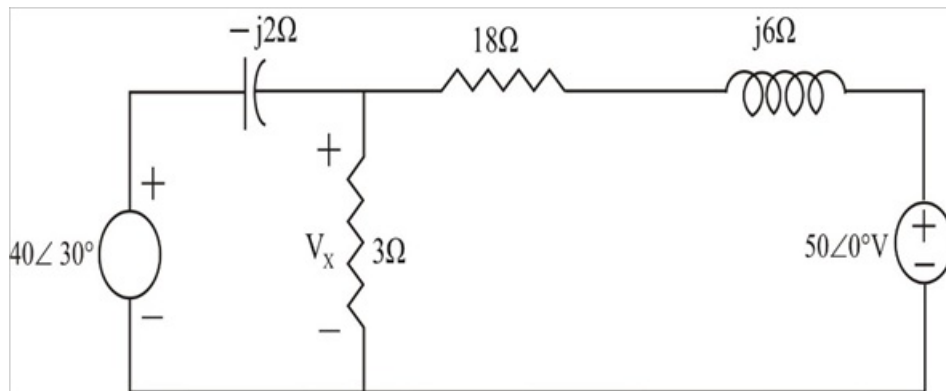
Figure 10.62:



Step-by-step solution

Step 1 of 3

Nodal analysis is the best approach to use on this problem. We can make our work easier by doing a source transformation on the right-hand side of the circuit.



Step 2 of 3

Figure 1

Step 3 of 3

$$\frac{V_x - 40\angle 30^\circ}{-j2} + \frac{V_x}{3} + \frac{V_x - 50}{18 + j6} = 0$$

$$V_x \left[\frac{1}{-j2} + \frac{1}{3} + \frac{1}{18 + j6} \right] - \frac{40\angle 30^\circ}{-j2} - \frac{50}{18 + j6} = 0$$

$$V_x \left[\frac{j}{2} + \frac{1}{3} + \frac{1}{18 + j6} \right] - \frac{40j\angle 30^\circ}{2} - \frac{50}{18 + j6} = 0$$

$$V_x \left[\frac{1\angle 90^\circ}{2} + \frac{1}{3} + \frac{1}{18 + j6} \right] - \frac{40\angle 30^\circ \times 1\angle 90^\circ}{2} - \frac{50}{18 + j6} = 0$$

$$V_x \left[\frac{1\angle 90^\circ}{2} + \frac{1}{3} + \frac{1}{18.97\angle 18.43^\circ} \right] - \frac{40\angle 120^\circ}{2} - \frac{50}{18.97\angle 18.43^\circ} = 0$$

$$V_x \left[\frac{1\angle 90^\circ}{2} + \frac{1}{3} + 0.0527\angle -18.43^\circ \right] - \frac{40\angle 120^\circ}{2} - 50 \times 0.0527\angle -18.43^\circ = 0$$

$$\frac{V_x (3\angle 90^\circ + 2 + 6 \times 0.0527\angle -18.43^\circ) - 3 \times 40\angle 120^\circ - 6 \times 2.635\angle -18.43^\circ}{6} = 0$$

$$V_x (3\angle 90^\circ + 2 + 0.3162\angle -18.43^\circ) - 120\angle 120^\circ - 15.81\angle -18.43^\circ = 0$$

$$V_x (2.9155\angle 84.094^\circ + 2) + 108.68\angle 114.46^\circ = 0$$

$$V_x (2.9155\angle 84.094^\circ + 2) = -108.68\angle 114.46^\circ$$

$$V_x = -\frac{108.68\angle 114.46^\circ}{3.70\angle 51.58^\circ}$$

$$= 29.37\angle 62.88^\circ \text{ V}$$

Hence, the voltage V_x is $\boxed{29.37\angle 62.88^\circ \text{ V}}$.